

Internet of Things (IoT)	Supervisory Control and Data Acquisition (SCADA)
”Servers”	Hardware
Motherboard	CPU
Memory	Hard Drive
GPU	Network Interface Card (NIC)

Systems that utilize simple computers in industrial applications (e.g. monitoring water treatment)	Traditional things (e.g. a refrigerator) have computers put in them
The physical computer	• May refer to "big, powerful computer" = many interconnected computers • May refer to software running on any computer whose job it is to deliver information to another computer or application on the same computer
• Central processing unit, core processing electronics of the computer, plugs into motherboard • Where processing/computation happens (the brain of the computer)	• Large silicone chip that connects other components, sets clock speed • Backbone of computer
Large, permanent information store; plugs in or tethers to motherboard; may be solid state (no moving parts) or spinning magnetic disks	• Small, volatile (temporary), fast information store near CPU; plugs into motherboard; comes in "sticks" (e.g. 4GB) • Typically RAM
• Used to interconnect computers • NOTE: Wired connection is often called ethernet	Graphics processing unit, secondary electronics originally intended for massively parallel computations for computer graphics, now/also deep learning

kodiak	Software
Firmware	Free Software (FOSS)
GNU Project	Gratis
Assembly	Compiled Language
Interpreted Language	Operating Systems (OSes)

A collection of data or computer instructions that tell the computer how to work.	Baylor's high performance cluster
Computer software distributed under terms that allow users to run the software for any purpose as well as to study, change, and distribute it and any adapted versions.	• Low-level software to provide a base layer for interacting with it • NOTE: you can kill a computer permanently if you interrupt firmware updates
No-cost software	Suite of free software with the following essential freedoms: 1. To run programs for any purpose 2. To study and change source code 3. To redistribute copies 4. To distribute copies of your modified source code
• Programming language that uses another program (the compiler) to convert source code into an executable program • Examples: C^{++}, Java • Compilation takes time, but only needs to be done once; compiled code is fast	Low-level programming language that mimics the core instructions that the hardware is designed with
Software that sit between user applications and the computers physical hardware Three main Operating Systems: • Windows • Mac • Linux	• Programming language that uses another program (the interpreter) to execute instructions directly • Examples: R, Python • Tends to do certain things slower (e.g. looping)

Major Linux Distributions	Graphical User Interface (GUI)
Terminal	Terminal Emulator
Shell	Virtualization
Hypervisor	NOTE:
Scalability	Containerization

Often use windows, mice, and point and click interfaces; touch interfaces	• Debian - a bit bare bones but a true believer in the free software movement • Ubuntu - downstream Debian, most common OS, commercialized • Red Hat Enterprise Linux (RHEL) - rock solid enterprise OS, paid support • CentOS - free downstream of RHEL, OS on kodiak
Software that displays a terminal (monitor) in a windowed system	Primary set of IO devices, i.e. keyboard and monitor
• Allows you to run a guest OS inside your own host OS • While resource heavy, it is not emulated by your OS; your CPU supports virtualization • Allows you to use a mostly sandboxed (isolated) OS, see what other OSes see	• A command-line interpreter which exposes access to an operating system's services • Program that sits between you and the OS and interprets its commands • Most popular: Bourne Again Shell (BASH)
• RAM is volatile memory • ROM is non-volatile memory	Software that enables virtualization
• Can be thought of as extremely lightweight virtualization • Major containerization platform is Docker	• Foundational aspect of cloud computing • Scaling up = Increases virtual resources of the virtual machine (VM) • Scaling out = Replicating/cloning virtual machines for cloning of many of the same tasks.

Kubernetes	Image
Cloud Computing	

A virtual computer or container that has been saved to a particular state and distributed to other people	Platform for orchestration of tons of containers
	Computations done on remote resources “in the cloud”